

Some useful references related to INLA

The Integrated Nested Laplace Approximation (INLA), **Rue et. al. (2009)** takes advantage of certain specification for the random effects to allow fast and flexible inference as illustrated in **Rue et. al. (2017)**. The specification is for the parameters in the linear predictor to have a joint Gaussian prior specified as a Gaussian Markov Random Field – GMRF. About GMRF see **Rue and Held (2005)**, or the and summarized text in **Rue and Held (2010)**. This allows for computational advantages when applying the Laplace method to integrate out these parameters, the main Laplace approximation by applying a Gaussian approximation as described in Chapter 4 of **Rue and Held (2005)**. The marginal inference for these parameters is later performed by applying the Laplace method as nested Laplace approximations, one for each of these parameters. The current state of the art replaces the nested Laplace approximations by a low-rank implicit variational Bayes correction to the main Laplace approximation, **van Niekerk and Rue, 2024**. It allowed a re-formulation of the original implementation that greatly improves speed as illustrated in **van Niekerk et. al. (2023)**, with no loose of accuracy.

One basic step in Bayesian inference is to assign prior for the parameters. For those in the linear predictor, except for sharing parameters, are assumed Gaussian, one for each model component and, thus, a joint Gaussian, as mentioned earlier. The other model parameters, named as hyper-parameters, are them parameters controlling the random effects or the likelihood extra parameters. For example, the default in INLA is to assume a gamma for precision parameters. However a new framework was proposed, **Simpson et. al. (2017)**, and many models in INLA have such a prior defined, where random effects can be understood as a more complex model than a simpler, where no random effect is present. Under this framework the prior is set considering the Kullback-Leibler divergence from the random effect distribution to the base distribution, where the random effect has zero variance, for example. Some correlation parameters follows similar idea. This prior is independent of model parametrization and the prior for a parameter is derived from a change of variables.

For the intrinsic random effects, such as the random walk models (for example for model = 'rw1', 'rw2', 'besag' in INLA), the marginal precision is achieved after model scaling, as recommended in **Sørbye S. and Rue, H. (2014)** and also for the scaled and reparametrized BYM model (model = 'bym2'), see **Riebler, A. et. al. (2016)**.

For areal spatio-temporal models, a key work is **Knorr-Held (2000)**, where the disease risk is spacetime inseparable but the prior distribution has a separable precision matrix for the spacetime interaction effect. Its scaled version can be implemented in INLA using the `inla.knmodels()` or with Kronecker of two 'generic0' models with the INLAtools package <https://github.com/eliaskrainski/INLAtools>

The automatic group cross validation was proposed by **Liu, Z. et. al. (2025)** Liu and is illustrated for different structured random effects in **Adin, A. et. al. (2024)**.

Links:

<https://www.r-inla.org/> <https://pyinla.org/>
<https://github.com/eliaskrainski/INLAspacetime>

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